

STUDENT ID CARD MANAGEMENT PORTAL USING AWS

SUBTITLE : SCALABLE , SECURE , CLOUD – BASED SOLUTION

PRESENTED BY

G HARISH & TEAM

258865100016

PROBLEM STATEMENT

- IN MANY COLLEGES, STUDENT ID CARD REQUESTS ARE HANDLED MANUALLY.
- STUDENTS HAVE TO VISIT THE OFFICE MULTIPLE TIMES.
- STAFF ALSO HAS DIFFICULTY MAINTAINING RECORDS.
- TO SOLVE THIS PROBLEM, WE DEVELOPED AN ONLINE CLOUD-BASED SYSTEM WHERE STUDENTS CAN APPLY FOR ID CARDS, TRACK STATUS, AND REPORT LOST ID CARDS.

OBJECTIVES

- TO AUTOMATE THE ID CARD REQUEST PROCESS.
- TO REDUCE MANUAL WORK.
- TO SECURELY STORE STUDENT INFORMATION IN THE CLOUD.
- TO MAKE THE SYSTEM ACCESSIBLE FROM ANYWHERE.

TECHNOLOGIES USED

Frontend:

JSP, HTML, CSS

Backend:

Java Servlets

Database:

MySQL

Server:

Apache Tomcat

Cloud Platform:

AWS

AWS SERVICES USED

- EC2
- RDS
- S3
- CLOUDWATCH
- VPC

Amazon EC2

- Amazon EC2 hosts our web application.
- Tomcat server is installed inside EC2.
- Users access the website through the EC2 Public IP

The screenshot displays the Amazon EC2 console interface. The top navigation bar includes the AWS logo, a search bar, and account information for 'United States (N. Virginia)' with Account ID: 1933-1418-2820. The left sidebar shows the 'EC2' menu with options like Dashboard, AWS Global View, Events, and a list of instances. The main content area is titled 'Instances (1/1)' and shows a single instance named 'Smart-ID-Card-Server' with ID 'i-080f289d24658d04a'. The instance is in a 'Running' state, using a 't3.micro' instance type, and is located in the 'us-east-1c' availability zone. Below the instance list, the 'Instance summary' section provides detailed information:

Instance summary	Public IPv4 address	Private IPv4 addresses
Instance ID i-080f289d24658d04a	3.227.57.98 open address	172.31.23.39
IPv6 address -	Instance state Running	Public DNS ec2-3-227-57-98.compute-1.amazonaws.com open address
Hostname type IP name: ip-172-31-23-39.ec2.internal	Private IP DNS name (IPv4 only) ip-172-31-23-39.ec2.internal	Elastic IP addresses 3.227.57.98 [Public IP]
Answer private resource DNS name IPv4 (A)	Instance type t3.micro	

The bottom of the console shows a footer with 'CloudShell', 'Feedback', and copyright information for Amazon Web Services, Inc. or its affiliates.

Amazon RDS

- Amazon RDS stores our MySQL database.
- User registration, login details, ID requests and Lost ID requests are stored in RDS.

The screenshot displays the AWS Management Console interface for the Amazon RDS console. The browser address bar shows the URL: `us-east-1.console.aws.amazon.com/rds/home?region=us-east-1#database:id=mca-project-db;is-cluster=false`. The console header includes the AWS logo, a search bar, and the account ID: 1933-1418-2820. The left sidebar shows the navigation menu with 'Aurora and RDS' selected, and 'Databases' is the active view. The main content area shows the details for the 'mca-project-db' database instance. The 'Summary' section includes the following information:

DB identifier	Status	Role	Engine	Recommendations
mca-project-db	Available	Instance	MySQL Community	2 Informational

Additional summary information includes:

CPU	Class	Current activity	Region & AZ
4.11%	db.t4g.micro	0 Connections	us-east-1c

The 'Connectivity & security' tab is selected, showing options for connecting to the database:

- Code snippets** (Selected): Use when connecting through SDK, APIs, or third-party tools including agents.
- CloudShell**: Use for a quick access to AWS CLI that launches directly from the AWS Management Console.
- Endpoints**: Use when connecting through any IDE interface.

Other settings shown include:

- Internet access gateway**: Disabled
- IAM Authentication**: Disabled
- Programming language**: MySQL (macOS)
- Endpoint type**: Instance endpoint

The 'Connection steps' section provides instructions for connecting to the database using the provided snippets.

Amazon S3

- Amazon S3
- stores files like Student
- Photos ID Card
- PDFs Documents

The screenshot shows the Amazon S3 console interface. The left sidebar contains navigation options: Amazon S3, Buckets, Files, Access management and security, Storage management and insights, and Account and organization. The main content area is titled 'Buckets' and shows 'General purpose buckets (1)'. A table lists the bucket 'mca-project-harish-2026' in the 'US East (N. Virginia) us-east-1' region, created on 'July 12, 2026, 23:50:51 (UTC+05:30)'. The right sidebar contains 'Account snapshot' and 'External access summary' sections. The bottom of the page includes a footer with 'CloudShell', 'Feedback', and copyright information.

Launch AWS Academy Learner | S3 buckets | S3 | us-east-1 | EC2 Instance Connect | us-east-1 | Ask Gemini

us-east-1.console.aws.amazon.com/s3/buckets?region=us-east-1

Search [Alt+S] United States (N. Virginia) Account ID: 1933-1418-2820 voclabs/user5065999=harishhari20004@gmail.com

Amazon S3 > Buckets

Amazon S3

- ▼ Buckets
 - General purpose buckets
 - Directory buckets
 - Table buckets
 - Vector buckets
- ▼ Files
 - File systems
- ▼ Access management and security
 - Access Points
 - Access Points for FSx
 - Access Grants
 - IAM Access Analyzer
- ▼ Storage management and insights
 - Storage Lens
 - Batch Operations
- Account and organization

Buckets

General purpose buckets All AWS Regions Directory buckets

General purpose buckets (1) Info

Refresh Copy ARN Empty Delete Create bucket

Buckets are containers for data stored in S3.

Find buckets by name

Name	AWS Region	Creation date
mca-project-harish-2026	US East (N. Virginia) us-east-1	July 12, 2026, 23:50:51 (UTC+05:30)

Account snapshot Info

Updated daily

View dashboard

Storage Lens provides visibility into storage usage and activity trends.

External access summary Info

Updated daily

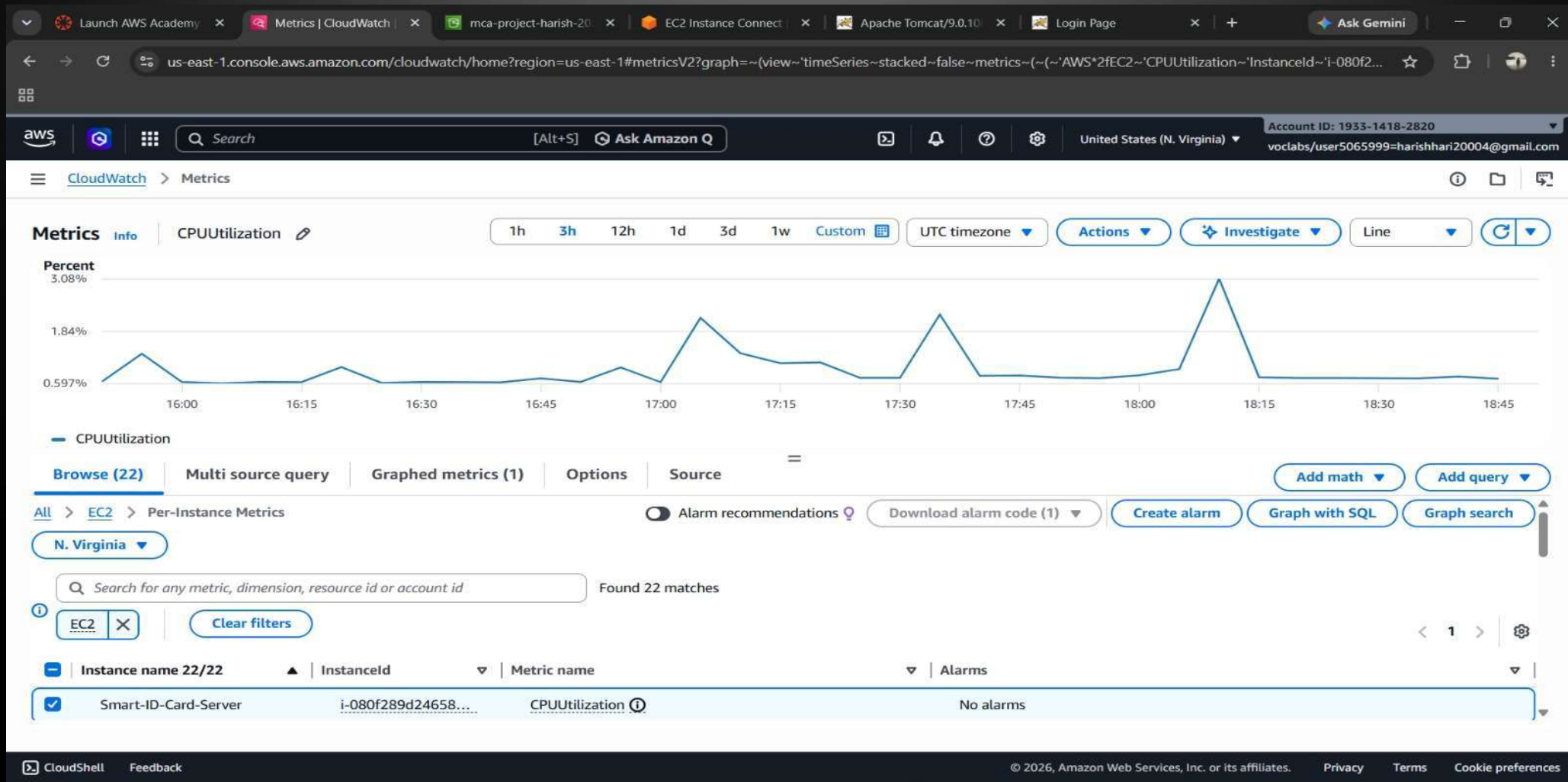
External access findings help you identify bucket permissions that allow public access or access from other AWS accounts.

CloudShell Feedback

© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

CloudWatch

- CloudWatch monitors our AWS resources.
- It monitors EC2
- CPU Memory Network RDS performance



VPC

- Provides a secure virtual network.
- Isolates cloud resources.
- Enables secure communication between AWS services

The screenshot displays the AWS VPC console interface. The top navigation bar includes the AWS logo, a search bar, and the region 'United States (N. Virginia)'. The main content area is titled 'Your VPCs' and shows a table of VPCs. A single VPC, 'smart-id', is listed with ID 'vpc-091306915e5912e18' and state 'Available'. Below the table, the 'Details' tab for this VPC is expanded, showing various configuration parameters.

Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv...
smart-id	vpc-091306915e5912e18	Available	-	-	Off	172

vpc-091306915e5912e18 / smart-id

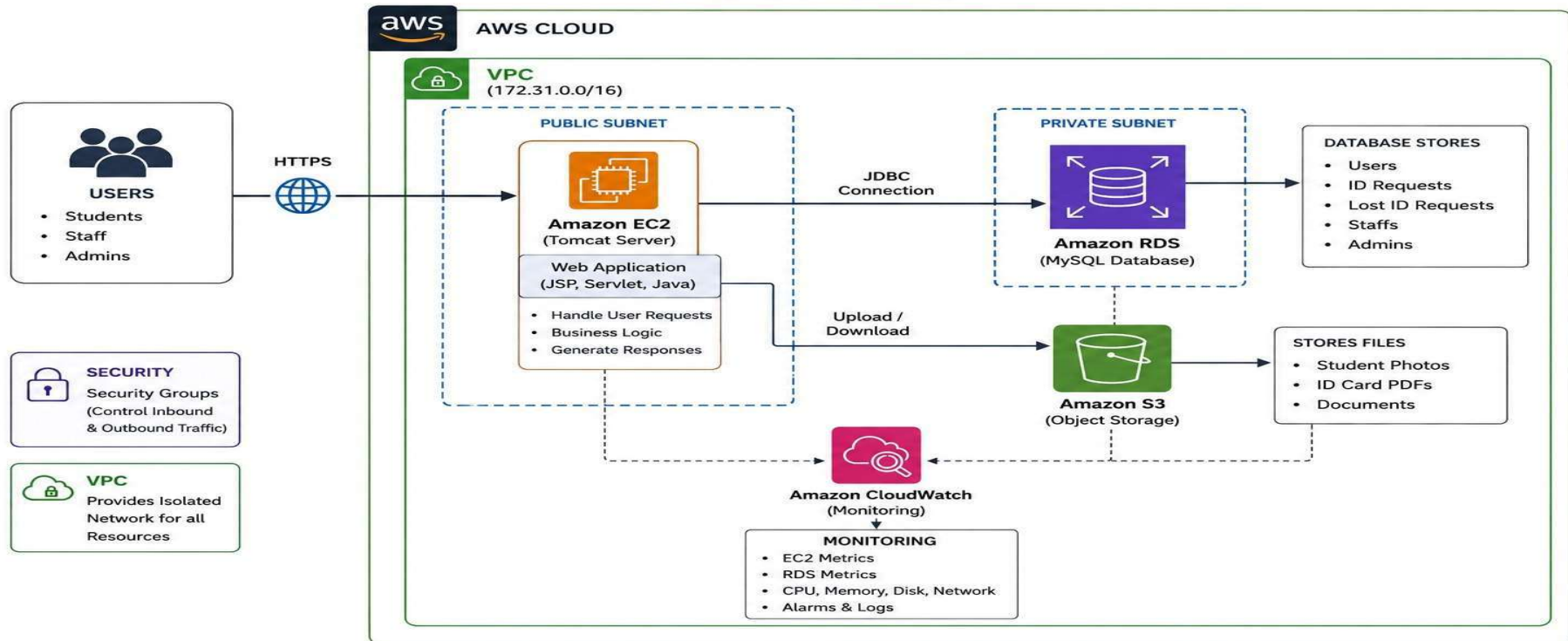
Details

VPC ID vpc-091306915e5912e18	State Available	Block Public Access Off	DNS hostnames Enabled
DNS resolution Enabled	Tenancy default	DHCP option set dopt-07d687028615fbde0	Main route table rtb-034fb4a99f6d37b65
Main network ACL acl-041bc79808cbd9f59	Default VPC Yes	IPv4 CIDR 172.31.0.0/16	IPv6 pool -
IPv6 CIDR (Network border group) -	Network Address Usage metrics Disabled	Route 53 Resolver DNS Firewall rule groups -	Owner ID 193314182820

ARCHITECTURE

- This is our system architecture.
- First, the Student opens our website.
- The request goes to Amazon EC2.
- Tomcat server processes the request.
- If database access is required, EC2 communicates with Amazon RDS.
- Student photos and documents are stored in Amazon S3.
- CloudWatch monitors the entire application.
- Security Groups provide secure communication

CLOUD-BASED STUDENT ID CARD MANAGEMENT SYSTEM AWS ARCHITECTURE



ADVANTAGES

- LESS PAPERWORK
- FASTER APPROVAL
- CLOUD STORAGE
- SECURE DATA
- EASY ACCESS
- LOW MAINTENANCE

FUTURESCOPE

- FUTURE ENHANCEMENTS:
- QR CODE ID CARDS
- FACE RECOGNITION
- MOBILE APPLICATION
- EMAIL NOTIFICATIONS
- AI-BASED VERIFICATION

CONCLUSION

- WE SUCCESSFULLY DEVELOPED A CLOUD-BASED STUDENT ID CARD MANAGEMENT SYSTEM USING AWS.
- THE PROJECT REDUCES MANUAL WORK AND SECURELY STORES DATA IN THE CLOUD.

The image features a dark gray background with several realistic, glossy bubbles of varying sizes. These bubbles are clustered in the top-left and bottom-right corners, creating a decorative frame. The central text 'THANK YOU' is rendered in a white, elegant serif font with a subtle drop shadow, giving it a three-dimensional appearance.

THANK YOU